

## Chapter 2

### REVIEW OF RELATED LITERATURE

#### 2.1 Agile Scrum Methodology Overview

Agile Scrum is chosen as the development methodology for the Legislative Research and Analysis Records Management System because of its adaptability and incremental development process, which effectively supports the integration of AI-powered research functions, analytical features, and centralized legislative data. The system is developed through a series of short development cycles, allowing each component such as document archiving, research source organization, intelligent search capabilities, and analysis reporting to be tested and refined independently before being combined into the full platform. This approach promotes regular evaluation and collaboration with Valenzuela LGU stakeholders, including **Mr. JoJo Faustino**, enabling early improvements that ensure the system resolves research workflow gaps, improves access to information, and strengthens data-driven legislative decision-making before final implementation.

#### 2.2 Enterprise Architecture Concepts

The Research and Analysis Legislative Records Management System is conceived as a strategic initiative as it plays a vital role in supporting evidence-based legislation within the Valenzuela City government. Beyond managing legislative records, the system is designed to facilitate research, analysis, and

informed decision-making processes. Achieving these objectives requires close alignment with the legislative department's organizational structure, workflows, and governance goals. In this context, Enterprise Architecture (EA) serves as a guiding framework that justifies the centralized integration of legislative records, research data, and analytical functions into a unified and coherent system.

## **2.3 Microservices Architecture**

This is for scalability, flexibility, and efficiency. Each service is responsible for specific functions: document handling, user access, and AI search, each running independently with the use of secured APIs. It enhances stability, enables faster updates, and keeps the processes in the Legislative Department of Valenzuela City running smoothly.

## **2.4 DevOps and CI/CD**

DevOps and Continuous Integration/Continuous Deployment (CI/CD) are approaches that make software development faster, safer, and more reliable. For the Legislative Record Management System (LRMS) at Valenzuela City Hall, these practices help the team develop efficiently, maintain the system, and quickly fix any issues.

### **2.4.1 DevOps**

DevOps promotes teamwork between Development and Operations teams. It focuses on:

**Automation:** Streamlining tasks like building, testing, and deploying the system.

**Rapid Feedback:** Quickly detecting errors or performance issues.

**Continuous Improvement:** Constantly refining the software delivery process.

#### **2.4.2 Continuous Integration (CI)**

CI means frequently merging code into a shared system, with automatic tests to catch errors early. For LRMS, this ensures that new module updates work well with existing code, preventing delays.

#### **2.4.3 Continuous Delivery (CD)**

CD ensures the software is always ready for deployment. For LRMS, this allows updates to go safely to testing environments for staff feedback (User Acceptance Testing) and ensures bug fixes can be applied without disrupting legislative work.

Using DevOps and CI/CD shows that the LRMS is not only being built effectively but is also designed to run reliably and scale within the city government system.

### **2.5 Relevant Studies and Research**

This chapter presents a review of relevant literature and studies that provide a foundation for understanding the significance of integrating technology into **local legislative processes**, particularly concerning **record management, digital transformation in government, and the use of AI and microservices architecture to improve information systems**. The review is divided into foreign

and local studies, highlighting how digital innovations and advanced architectural concepts contribute to more **efficient, secure, and data-driven legislative governance**.

## **Local Studies**

In the Philippines, the Data Privacy Act of 2012. The proposed policy aims to strengthen operational efficiency, legal compliance, and transparency in LGU service delivery. This law provides the legal framework for the security requirements (hashing, RBAC, etc.) of the LIS, specifically the need for secured digital archives.

**Tabia (2025)**, many LGUs still rely on manual and fragmented methods for handling legislative documents, resulting in delayed access, data duplication, and challenges in maintaining accurate records. The study emphasizes that digital transformation is essential for achieving transparency, accountability, and faster service delivery in government offices.

*Tabia, M. (2025). Digital transformation in local government units: Addressing challenges in legislative document management. International Journal for Multidisciplinary Research (IJFMR). Retrieved from: <https://www.ijfmr.com/research-paper.php?id=45853>*

**Parreño, Tudara, and Soberano (2024)** investigated the development and implementation of a **Legislative Data Archiving and Records Keeping System (LEDARKS)** in the Sangguniang Bayan Office of Hinoba-an. Their study highlights

the limitations of manual record-keeping, including inefficiencies, difficulty in retrieval, and potential data loss, and emphasizes how an organized digital repository with standardized metadata and centralized access significantly improves legislative document management and accessibility. The research also noted key considerations for effective implementation, such as data security, interoperability, and staff training.

*Parreño, J. D., Tudara, J. M., & Soberano, K. (2024). Legislative Data Archiving and Records Keeping System.*

**Libadia, Botangen, Lucero, and Tecson (2025)** developed a **web-based Electronic Records Management System (ERMS)** for the Office of Senior Citizen Affairs in the Philippines, aiming to centralize record-keeping and streamline the storage, processing, and retrieval of records. Their study highlighted how a web-based ERMS addresses inefficiencies associated with manual record management by improving data organization, real-time monitoring, report generation, and overall workflow efficiency, demonstrating the system's usability, reliability, and security in practical application.

*Libadia, M., Botangen, K. A., Lucero, P. J., & Tecson, J. (2025). Development of a Web-based records management system: an ERMS initiative for the Office of Senior Citizen Affairs in the Philippines. Records Management Journal, 35(2), 160-179.*

**Manun-og and Flores (2024)** developed a municipal consolidation system for the registry of inhabitants' records to optimize local governance record management in Hinunangan, Southern Leyte. The study found that the system—designed using an iterative waterfall model and evaluated through user testing—achieved high usability, acceptance, and improved centralized access to inhabitants' data. The authors concluded that adopting such technology-driven solutions can enhance record retrieval, reporting, and overall efficiency in local government units.

*Manun-Og, M. B., & Flores, J. B. (2024). Optimizing registry of inhabitants' record management: the development of municipal consolidation system for local governance. SciEnggJ, 17, 8-17.*

**Kotyla (2021)** examined the use of **Electronic Document Management Systems (EDMS)** in Polish local government units to process accounting electronic documents. The study used survey research across numerous LGUs and found that while a majority of units had implemented EDMS tools, many were still not fully prepared for electronic document registration and processing, particularly for structured formats needed for automation. The findings suggest that more comprehensive adoption and integration of digital systems are needed to fully support e-accounting and e-document workflows in local governance.

*Kotyla, C. (2021). Electronic Document Management Systems as an IT tool for processing accounting e-documents in Polish local government. Zeszyty Teoretyczne Rachunkowości, (111), 112-134.*

## **Foreign Studies**

Gunnlaugsdóttir (2012) investigated how **information and records management (IRM)** acts as a foundational element for effective quality management systems within organizations. Based on a survey of Icelandic firms certified under the ISO 9001:2008 quality standard, the study found that strong IRM contributed to improved management oversight, disciplined work procedures, enhanced transparency, and better compliance with customer and regulatory requirements. The research highlights that shortcomings in IRM can hinder the achievement and maintenance of quality certifications, emphasizing the strategic value of systematic records practices in organizational performance.

*Sakinat, J. A., MOHAMMED, R., & AHMED, J. (2024). USE OF ELECTRONIC RECORD MANAGEMENT SYSTEM FOR EFFECTIVE PERFORMANCE BY REGISTRY STAFF OF NATIONAL BOARD FOR ARABIC AND ISLAMIC STUDIES. NASARAWA JOURNAL OF LIBRARY AND INFORMATION SCIENCE (NAJLIS), 8(1), 277-287.*

**Burkan Hawash (2021)** — The study “*Electronic records management systems and the competency of educational institutions: Evidence from Yemen*” explores how **Electronic Records Management Systems (ERMS)** influence the

ability of public-sector educational institutions to manage records and information effectively. It proposes a framework for ERMS adoption and identifies **critical factors** — including ERMS characteristics and cloud technology features — that support successful implementation. Findings suggest that a well-implemented ERMS enhances institutional competency, regulatory compliance, and operational performance by improving accessibility, organization, and management of electronic records. This research contributes empirical evidence on ERMS adoption and highlights the strategic role of electronic recordkeeping in strengthening organizational functions.

*Mukred, M., Yusof, Z. M., Al-Moallemi, W. A., Mokhtar, U. A. A., & Hawash, B. (2022). Electronic records management systems and the competency of educational institutions: Evidence from Yemen. Information Development, 38(1), 125-148.*

**Udensi and Akor (2014)** conducted a study assessing the **record management system in the establishment divisions of two Nigerian universities**—Federal University of Technology Minna and Ibrahim Badamasi Babangida University, Lapai. Their research revealed that record management practices were largely ineffective due to inadequate infrastructure, insufficient trained staff, and persistent manual processes, which hinder efficient access, retrieval, and preservation of records. The authors recommended improved training, better facilities, and systematic record management policies to enhance effectiveness and support administrative functions.



*Akor, P. U., & Udensi, J. (2013). An assessment of record management system in establishment division of two universities in Nigeria. International Letters of Social and Humanistic Sciences, 13(2), 97-109.*

## **2.6 Integration of Information Systems in Enterprise Environment**

In an enterprise environment, a **Legislative Records Management System (LRMS)** acts as a centralized hub that integrates various information systems to ensure the legal integrity, transparency, and accessibility of the legislative process. This integration transforms a simple document repository into a sophisticated enterprise ecosystem where data flows seamlessly between lawmakers, administrative staff, and the public.

### **Enterprise Integration Framework**

The LRMS is built upon a multi-layered architecture designed to eliminate "information silos." At the **Data Integration** layer, the system synchronizes diverse inputs—such as bill drafts, committee meeting minutes, and official voting records—into a "Single Source of Truth." This ensures that when a staff member updates a bill's status, that change is immediately reflected on the public-facing **Dashboard**, preventing version conflicts that could lead to legal or procedural errors.

## Operational Workflow and Role Alignment

The integration of these systems is governed by strict **Identity and Access Management (IAM)** protocols, which define the responsibilities of each actor within the enterprise:

- **Administrators:** They manage the high-level infrastructure, ensuring that **Audit Logs** are immutable and secure. In a legislative context, this provides a "defensible record" of every amendment made to a law, fulfilling legal requirements for accountability.
- **Staff (Clerks and Researchers):** They utilize integrated **Research and Analysis** tools to pull historical data into current drafts. The enterprise environment allows them to move documents through a digital "workflow engine," where a bill automatically proceeds from drafting to committee for review based on predefined business rules.
- **Citizens:** Through a secure portal, the public interacts with the system to **View and Download Documents**. The integration of an **AI Chat Assistant** serves as a **real-time interactive guide**, providing users with instant step-by-step navigation instructions and troubleshooting support to ensure they can easily use every feature within the system.

## Systemic Benefits and Strategic Value

By integrating these functional modules, a legislative body achieves **Operational Efficiency** and **Business Intelligence**. The system does not just store records; it generates **Reports and Analytics** that help leadership track the

pace of legislation and identify bottlenecks in the committee's process.

Furthermore, the use of **Enterprise Content Management (ECM)** ensures that these records are archived according to national standards, preserving the legislative history of the state for future generations.

This integrated approach ensures that the legislative process is not only digitized but also optimized for high-stakes decision-making and public trust.